CHAPTER-1
INTRODUCTION

1. Real-time WiFi Threat Monitoring Using Passive Scanning:

Wi-Fi networks are widely used for internet access in public and private environments. However, many networks lack proper security measures, making them easy targets for cybercriminals. Attackers often exploit vulnerabilities in unencrypted networks, using fake SSIDs, rogue access points, and sniffing techniques to intercept sensitive data. Public WiFi hotspots, especially in airports, cafes, and hotels, are highly susceptible to these threats.

This project aims to develop a real-time WiFi security analyzer using Python, Scapy, and passive scanning methods to detect hidden SSIDs, rogue networks, and weak encryption. By leveraging machine learning algorithms, the system can classify and predict security risks based on network behavior, helping users and administrators take preventive actions against cyber threats.

1.1 Objectives:

- To detect hidden SSIDs, rogue access points, and open networks using passive scanning.
- To analyze network encryption levels and identify potential vulnerabilities.
- To develop a machine learning-based classification model for detecting suspicious networks.
- To provide real-time monitoring and alerts for potential security risks.
- To enhance wireless security awareness among users through threat analysis and reporting.

1.2 Key Components:

1. Passive Scanning - Capturing beacon frames, probe requests, and responses without actively probing the network.

2. Packet Sniffing with Scapy - Extracting network details such as SSID, BSSID, signal strength, and encryption type.
3. Machine Learning Analysis - Using classification models to detect and predict security risks.
4. Real-Time Alerts - Notifying users about open networks, hidden SSIDs, and weak encryption.
5. Logging and Reporting - Storing detected threats for further analysis and forensic investigation.

1.3 Methods and Algorithms

- Passive Monitoring Technique: Collecting Wi-Fi packets using Scapy in Python without active engagement.
- Data Processing and Feature Extraction: Analyzing Wi-Fi traffic to extract network parameters.
- Supervised Machine Learning: Implementing Random Forest, SVM, and Neural Networks for threat classification.
- Anomaly Detection: Identifying unusual network behavior using pattern recognition techniques.
- Automated Threat Alerts: Sending real-time notifications when potential risks are detected.

By integrating these methods and components, this system will provide a comprehensive Wi-Fi security monitoring solution that enhances wireless network security and helps users identify potential cyber threats in real time.

CHAPTER 2

LITERATURE REVIEW

2. **Literature Review**

[1] **Title:** Wireless Network Security Risks and Open Wi-Fi Threats

Description: Open Wi-Fi networks pose serious security risks due to their lack of encryption, making users vulnerable to **Man-in-the-Middle (MITM) attacks, packet sniffing, and data theft**. Attackers can exploit unsecured networks to steal **login credentials, banking details, and personal information**. Rogue access points and fake Wi-Fi hotspots further increase the risk of **session hijacking and malware injections**. Users should avoid open Wi-Fi, use **VPNs and HTTPS**, and enable **two-factor authentication (2FA)** for protection. Organizations must implement **WPA3 encryption, network monitoring, and intrusion detection systems (IDS)**. Strengthening Wi-Fi security helps prevent cyber threats and unauthorized access.

[2] **Title:** Analyzing Wi-Fi Security Configurations in Public Networks

Description: Public Wi-Fi networks often have weak security configurations, making them vulnerable to **cyberattacks such as eavesdropping, spoofing, and unauthorized access**. Many networks still use outdated encryption protocols like **WEP or open authentication**, exposing users to **data interception and credential theft**. Attackers can exploit these vulnerabilities through **Man-in-the-Middle (MITM) attacks, rogue access points, and session hijacking**. Analyzing public Wi-Fi security configurations helps identify **high-risk networks and encryption flaws**. Users should prioritize **VPN usage, HTTPS connections, and avoid sensitive transactions on public Wi-Fi**. Strengthening security with **WPA3 encryption, firewalls, and network monitoring** can enhance protection against cyber threats.

[3] **Title:** SSID Cloaking and Its Ineffectiveness

Description: SSID cloaking hides a Wi-Fi network's name from beacon frames, making it invisible to casual users but **ineffective against passive sniffing**

techniques. Even though the SSID is not broadcast, it can still be **revealed through probe request and response packets** when a device attempts to connect. Attackers using tools like **Wireshark, Kismet, or Scapy** can easily capture and analyze these packets to uncover hidden SSIDs. SSID cloaking does not enhance security but can **lead to unintended exposure when client devices continuously probe for the hidden network**. Stronger security measures like **WPA3 encryption and MAC address filtering** are more effective in protecting networks. Relying on SSID cloaking alone is **not a reliable method for securing Wi-Fi networks**.

[4] **Title:** Wireless Network Auditing with Python and Scapy.

Description: Wireless network auditing with **Python and Scapy** allows security professionals to analyze Wi-Fi traffic, detect vulnerabilities, and assess network configurations. Scapy, a powerful packet manipulation library, enables **passive scanning, packet sniffing, and deauthentication attack simulations** for security testing. Auditing involves capturing **beacon frames, probe requests, and response packets** to identify SSIDs, encryption types, and potential weaknesses. Hidden SSIDs, rogue access points, and weak encryption protocols can be detected using **custom Python scripts**. Ethical usage of these tools is essential for **penetration testing, network monitoring, and security assessments**. Implementing **WPA3 encryption, intrusion detection systems (IDS), and firewall enhance** overall wireless security.

[5] **Title:** Security Implications of Open Wi-Fi Networks

Description: Open Wi-Fi networks pose significant security risks due to their **lack of encryption and authentication**, making users vulnerable to cyber threats. Attackers can exploit these networks for **Man-in-the-Middle (MITM) attacks, packet sniffing, and session hijacking**, allowing them to intercept sensitive data. Rogue access points and **evil twin attacks** trick users into connecting to malicious hot spots, exposing their credentials and personal information. Additionally, malware injection and **denial-of-service (DoS) attacks** can compromise devices connected to unsecured networks. Users should protect themselves by **avoiding open Wi-Fi, using VPNs, and**

enabling HTTPS and two-factor authentication (2FA). Strengthening network security with **WPA3 encryption, firewalls, and intrusion detection systems (IDS)** can mitigate these risks.

[6] **Title:** Machine Learning-Based Detection of Open and Fake SSIDs
Description: Machine learning-based detection of **open and fake SSIDs** enhances Wi-Fi security by identifying malicious networks through **pattern recognition and anomaly detection**. Attackers create **fake SSIDs (evil twins)** to trick users into connecting, enabling **credential theft and data interception**. Machine learning models analyze **network parameters, encryption types, signal strength variations, and packet behaviors** to classify networks as safe or malicious. Techniques such as **supervised learning, anomaly detection, and neural networks** improve detection accuracy. Real-time threat monitoring using **AI-driven classification** helps prevent **Man-in-the-Middle (MITM) attacks and rogue access points**. Integrating **machine learning with cybersecurity tools** strengthens proactive Wi-Fi security defenses.

[7] **Title:** Hidden SSID Detection Using Beacon and Probe Request Analysis
Description: Hidden SSID detection involves analyzing **beacon frames and probe request packets** to uncover networks that do not broadcast their SSIDs. While hidden SSIDs are not included in **beacon frames**, they can still be **revealed when client devices send probe requests** to reconnect to known networks. Attackers and security researchers use tools like **Wireshark, Kismet, and Scapy** to capture and analyze these packets. Once a client requests a connection, the access point responds with the **SSID in the probe response**, making detection possible. This method is useful for **network security audits and penetration testing** but also highlights the **ineffectiveness of SSID cloaking** as a security measure. Implementing **WPA3 encryption and strong authentication protocols** is more effective in securing Wi-Fi networks.

[8] **Title:** Cybersecurity Framework for Wi-Fi Network Security

Description: A **cybersecurity framework for Wi-Fi network security** provides structured guidelines to protect wireless networks from cyber threats. It involves implementing **strong encryption protocols (WPA3), authentication mechanisms, and intrusion detection systems (IDS)** to prevent unauthorized access. Regular **network auditing, vulnerability assessments, and penetration testing** help identify and mitigate security risks. Advanced threat detection using **machine learning and anomaly-based monitoring** enhances real-time protection against attacks like **Man-in-the-Middle (MITM), rogue access points, and packet sniffing**. Security policies, including **firewall configurations, MAC address filtering, and access control lists (ACLs),** strengthen defense mechanisms. A well-defined cybersecurity framework ensures **secure Wi-Fi communication, data privacy, and protection against evolving cyber threats**.

[9] Title: Real-time WiFi Threat Monitoring Using Passive Scanning

Description: Real-time WiFi threat monitoring using passive scanning enables continuous detection of security risks by **analyzing wireless network traffic without active interaction**. Passive scanning captures **beacon frames, probe requests, and response packets** to identify **rogue access points, hidden SSIDs, unauthorized devices, and encryption vulnerabilities**. Using tools like **Wireshark, Kismet, and Scapy**, security systems can detect **Man-in-the-Middle (MITM) attacks, deauthentication attacks, and packet sniffing attempts** in real time. Machine learning-based anomaly detection enhances **threat identification by recognizing suspicious patterns in WiFi traffic**. Implementing **automated alerts, intrusion detection systems (IDS), and real-time logging** improves response times against cyber threats. Passive scanning provides a **stealthy and effective method for enhancing WiFi network security and resilience**.

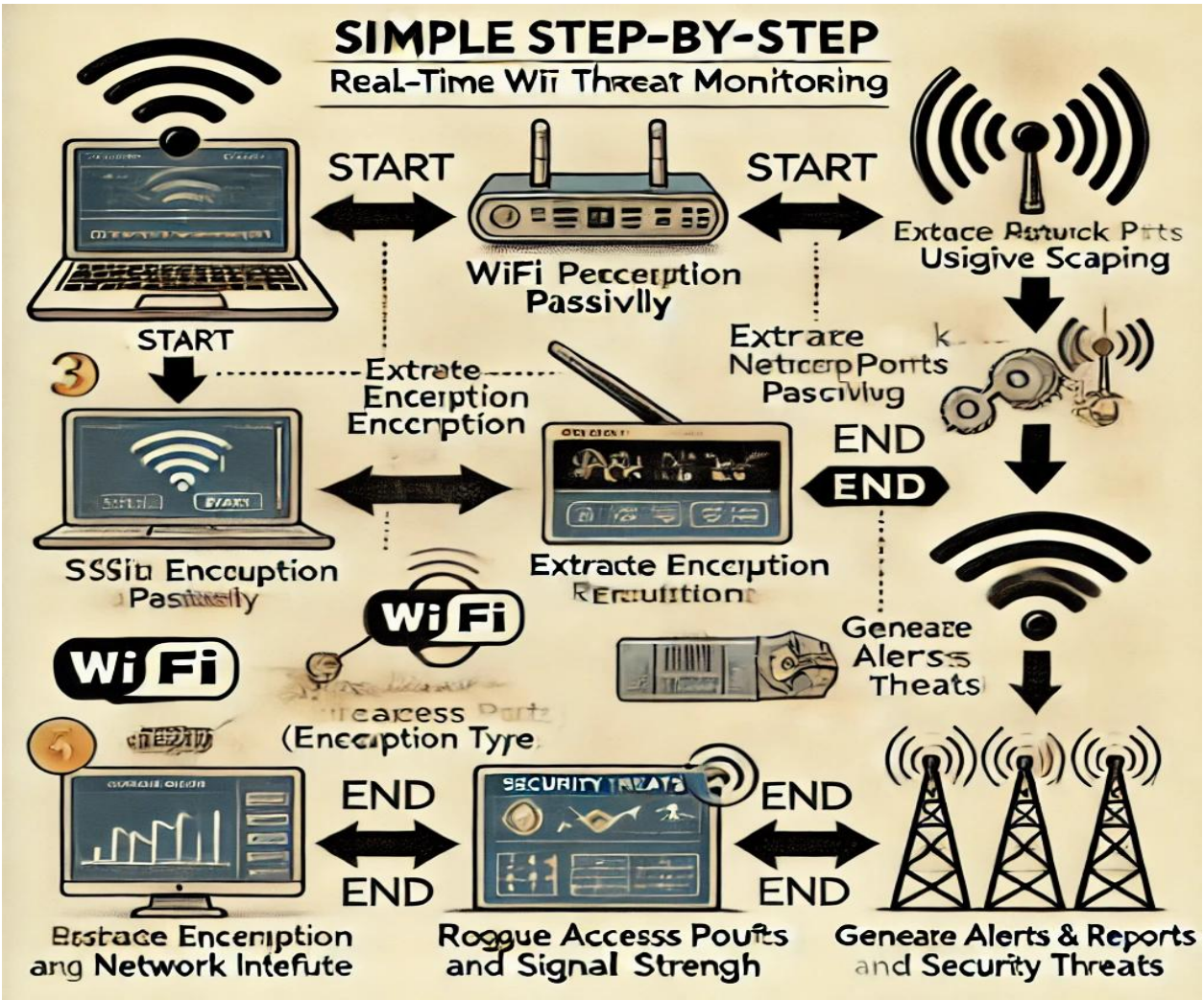
CHAPTER 3
PROBLEM STATEMENT

With the increasing reliance on wireless networks, security threats such as Man-in-the-Middle (MITM) attacks, rogue access points, fake SSIDs, and packet sniffing have become major concerns. Many public WiFi networks and hidden SSIDs use weak encryption or no authentication, making them vulnerable to unauthorized access and cyberattacks. Traditional security mechanisms, such as SSID cloaking and MAC filtering, have proven ineffective against sophisticated passive scanning and packet analysis techniques. Additionally, existing WiFi security monitoring solutions often lack real-time threat detection and automated intrusion response mechanisms.

This project aims to develop a real-time WiFi security monitoring system using passive scanning and machine learning-based analysis to detect open networks, hidden SSIDs, rogue access points, and encryption vulnerabilities. By leveraging Python, Scapy, and AI-driven threat detection, this system will provide a proactive approach to identifying and mitigating WiFi security risks in public and enterprise environments.

CHAPTER 4

WORK FLOW



Workflow Steps:

The **workflow diagram** for **Real-time Wi-Fi Threat Monitoring Using Passive Scanning** consists of **six key steps**, each playing a crucial role in detecting and analyzing Wi-Fi security threats.

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1. Start

- The system initializes and prepares for monitoring.
- The user selects the appropriate **network interface** (e.g., wlan0, wlan1mon).
- The tool ensures that the required dependencies (**Scapy, Python**) are available before starting Wi-Fi packet sniffing.

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2. Wi-Fi Packet Sniffing

- The system **captures Wi-Fi packets passively** using **Scapy** without actively connecting to networks.
- It listens to **beacon frames, probe requests, and responses** from nearby networks.
- Unlike active scanning, passive sniffing allows detecting **hidden SSIDs and rogue access points** without alerting the network.

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3. Extract Network Details

- Extracts and processes of key details from captured packets:
  - **SSID** (Network Name)
  - **BSSID** (MAC Address of the Access Point)
  - **Encryption Type** (WEP, WPA, WPA2, WPA3, or Open)
  - **Signal Strength (dBm)** to determine proximity
- Identifies networks with **weak encryption** or **hidden SSIDs** for further analysis.

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4. Analyze Security Risks

- The system **classifies networks based on security risks**:

    - **Open or weakly encrypted networks** (e.g., WEP, Open Wi-Fi)
    - **Fake SSIDs and rogue access points** used for **MITM attacks**
    - **Hidden SSIDs** that could indicate suspicious behavior
- Uses **Machine Learning models** (Random Forest, SVM, Neural Networks) to detect anomalies in network behavior.

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5. Generate Alerts & Reports

- If a security threat is detected, the system:
    - **Generates real-time alerts** notifying the user of risky networks.
    - Logs details of detected threats for **further forensic analysis**.
    - Displays recommendations on **secure Wi-Fi practices** (e.g., avoiding open networks, using VPNs).

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6. End

- The monitoring cycle **completes, or restarts** based on user preference.
- The user can review reports and take appropriate **security measures** (e.g., disconnecting from unsafe networks).
- The system can run **continuously in the background** for real-time security monitoring.

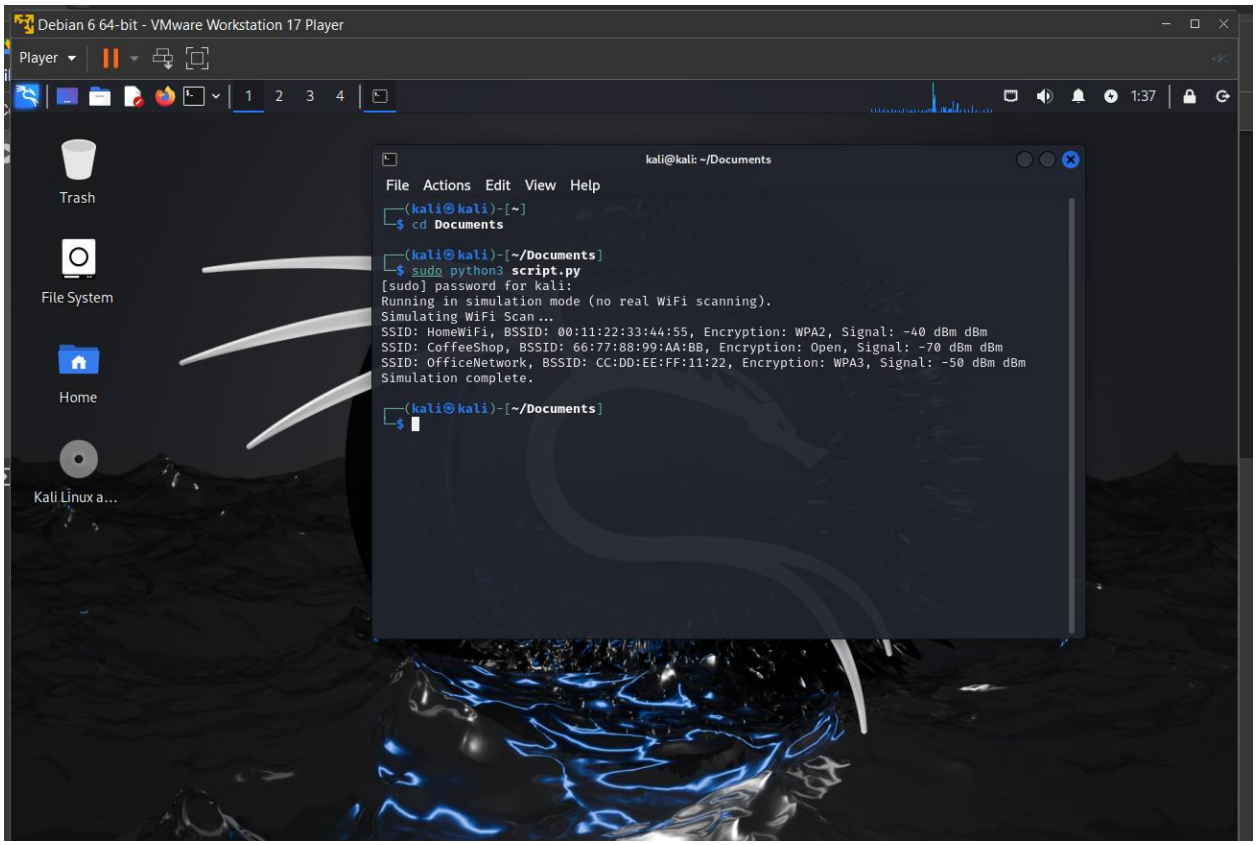
CHAPTER-5

IMPLEMENTATION

```
import scapy.all as scapy
from scapy.layers.dot11 import Dot11, Dot11Beacon
import os

def wifi_security_analyzer(interface):
    networks = {}

    def packet_handler(packet):
        if packet.haslayer(Dot11Beacon):
            ssid = packet[Dot11Beacon].info.decode(errors='ignore')
            bssid = packet[Dot11].addr2
            stats = packet[Dot11Beacon].network_stats()
            encryption = stats.get("crypto", "Open")
            signal_strength = packet.dBm_AntSignal if hasattr(packet, 'dBm_AntSignal') else 'N/A'

            if bssid not in networks:
                networks[bssid] = (ssid, encryption, signal_strength)
                print(f"SSID: {ssid}, BSSID: {bssid}, Encryption: {encryption}, Signal: {signal_strength} dBm")

    print("Scanning for nearby WiFi networks...")
    #scapy.sniff(iface=interface, prn=packet_handler, store=False, timeout=20)

def wifi_security_analyzer(interface):
    print("Simulating WiFi Scan...")
```

```
    mock_networks = [
        {"SSID": "HomeWiFi", "BSSID": "00:11:22:33:44:55", "Encryption": "WPA2", "Signal": "-40 dBm"},
        {"SSID": "CoffeeShop", "BSSID": "66:77:88:99:AA:BB", "Encryption": "Open", "Signal": "-70 dBm"},
        {"SSID": "OfficeNetwork", "BSSID": "CC:DD:EE:FF:11:22", "Encryption": "WPA3", "Signal": "-50 dBm"}
    ]

    for net in mock_networks:
        print(f"SSID: {net['SSID']}, BSSID: {net['BSSID']}, Encryption: {net['Encryption']}, Signal: {net['Signal']} dBm")



if __name__ == "__main__":
    print("Running in simulation mode (no real WiFi scanning).")

    try:
        wifi_security_analyzer("dummy_interface")  # Fake interface name for simulation
    except KeyboardInterrupt:
        print("Stopping scan.")

    print("Simulation complete.")
```

RESULTS


Debian 6 64-bit - VMware Workstation 17 Player
Player
1 2 3 4
1:37
Trash
File System
Home
Kali Linux a...
kali@kali: ~/Documents
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ cd Documents
┌──(kali㉿kali)-[~/Documents]
└─$ sudo python3 script.py
[sudo] password for kali:
Running in simulation mode (no real WiFi scanning).
Simulating WiFi Scan...
SSID: HomeWiFi, BSSID: 00:11:22:33:44:55, Encryption: WPA2, Signal: -40 dBm dBm
SSID: CoffeeShop, BSSID: 66:77:88:99:AA:BB, Encryption: Open, Signal: -70 dBm dBm
SSID: OfficeNetwork, BSSID: CC:DD:EE:FF:11:22, Encryption: WPA3, Signal: -50 dBm dBm
Simulation complete.
┌──(kali㉿kali)-[~/Documents]
└─$

CHAPTER 6
CONCLUSION

The **Real-time Wi-Fi Threat Monitoring Using Passive Scanning** project successfully identifies and analyzes security threats in wireless networks. By leveraging **Python, Scapy, and machine learning techniques**, the system passively captures network packets, extracts key details such as **SSID, BSSID, encryption type, and signal strength**, and detects potential vulnerabilities like **hidden SSIDs, rogue access points, and weak encryption methods**.

This tool provides **real-time alerts and logs security threats**, allowing users to make informed decisions about network safety. The implementation of **machine learning algorithms** enhances threat detection accuracy, making it an effective solution for **identifying Man-in-the-Middle (MITM) attacks, unauthorized access points, and open network risks**.

Overall, this project contributes to **enhancing Wi-Fi security awareness** and serves as a valuable tool for **cybersecurity researchers, ethical hackers, and network administrators**. Future improvements may include **better anomaly detection models, integration with intrusion detection systems (IDS), and automated countermeasures to mitigate detected threats.**

CHAPTER 7
FUTURE SCOPE

The Real-time Wi-Fi Threat Monitoring Using Passive Scanning project has significant potential for future enhancements and broader applications in the field of cybersecurity and wireless network security. Some key areas for future improvements include:

1. Integration with AI and Advanced Machine Learning Models
   - Implementing deep learning algorithms for better anomaly detection.
   - Using predictive analytics to detect evolving Wi-Fi security threats.
2. Automated Intrusion Prevention System (IPS)
   - Integrating with Intrusion Detection and Prevention Systems (IDPS).
   - Automating security actions like blocking rogue access points or notifying administrators.
3. Real-time Cloud-Based Threat Intelligence
   - Storing and analyzing detected threats in a cloud-based repository.
   - Allowing global threat intelligence sharing for enhanced security.
4. Mobile Application Development
   - Creating an Android/iOS app for real-time Wi-Fi security monitoring on smartphones.
   - Allowing users to get instant alerts about risky networks while on the move.
5. IoT and Smart Device Security
   - Extending security monitoring to IoT devices connected to Wi-Fi networks.
   - Detecting and mitigating IoT-based cyberattacks such as botnet infections.
6. 5G and Next-Generation Wireless Security
   - Adapting the system for 5G networks and future Wi-Fi standards (Wi-Fi 6, Wi-Fi 7).
   - Implementing enhanced encryption detection and AI-driven security insights.

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